**Goals:**

- Having a working automatic marker remover
- Improve data accuracy and trust in parking availability
- Enhance performance and scalability of the map experience
- Increase user insight

## User Story 3.1 (5 pts)

**As a user, I want to access an accurate, full history of my reported spots so that I can review my previous contributions.**

**Tasks:**

- Design Firestore query to fetch only reports created by the current user (1.5 hrs)
- Create “My Activity” screen layout (list view) in React Native (2 hrs)
- Display report metadata (location, spot type, timestamp, status) (1.5 hrs)
- Handle empty state and loading indicators (1 hr)

**Estimated Time:** 6 hrs

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## User Story 3.2 (5 pts)

**As a user, I want my locally saved parking spot reports to sync to the cloud without creating duplicates so that the map and my history stay accurate and uncluttered.**

**Tasks:**

- Identify duplicate-detection strategy (e.g., spot ID, location + timestamp window) (1.5 hrs)
- Implement client-side check before uploading local reports (2 hrs)
- Add Firestore safeguards (e.g., unique doc IDs or merge logic) (1.5 hrs)
- Test sync behavior with simulated offline → online scenarios (1 hr)

**Estimated Time:** 6 hrs

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## User Story 3.3 (5 pts)

**As a driver, I want the app to only load and display parking spots within the current map region so that the map stays fast and responsive while I pan and zoom.**

**Tasks:**

- Capture visible map bounds from react-native-maps region change events (1.5 hrs)
- Update Firestore query to fetch spots within bounding box (2 hrs)
- Throttle map movement queries to avoid excessive reads (1 hr)
- Test performance improvements with dense marker scenarios (1.5 hrs)

**Estimated Time:** 6 hrs

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## User Story 3.4 (13 pts)

**As a driver, I want my reported parking spot to be automatically marked as taken when I arrive so that other users aren't misled by outdated availability.**

**Tasks:**

- Track last reported spot locally (ID, location, timestamp) (1.5 hrs)
- Implement foreground location polling while app is open (2 hrs)
- Calculate proximity and dwell time logic (distance + time threshold) (2 hrs)
- Detect "parked" state using low movement or distance variance (2 hrs)
- Update Firestore report status to "TAKEN" when conditions are met (1.5 hrs)
- Display confirmation UI feedback when auto mark (2 hrs)

**Estimated Time:** 11 hrs

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## User Story 3.5 (8 pts)

**As a driver, I want to be able to undo an automatically marked “taken” parking spot within a short time window so that incorrect automatic updates do not reduce the accuracy of parking availability.**

**Tasks:**

- Design undo time window logic (e.g. 30–60 seconds) (1 hr)
- Implement UI notification with “Undo” action (2 hrs)
- Revert Firestore status update if Undo is selected (2 hrs)
- Prevent undo after time window expires (1.5 hrs)
- Test undo flow and edge cases (multiple taps, timeout) (1.5 hrs)

**Estimated Time:** 8 hrs

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**Initial Task Assignment:**

- Kenny: Research more on how to implement 3.4 + 3.5
- Ethan: Research more on how to implement 3.4 + 3.5
- Maxwell: Get started on 3.1
- Kyle: Get started on 3.2/ Add logout functionality

**Team Roles:**

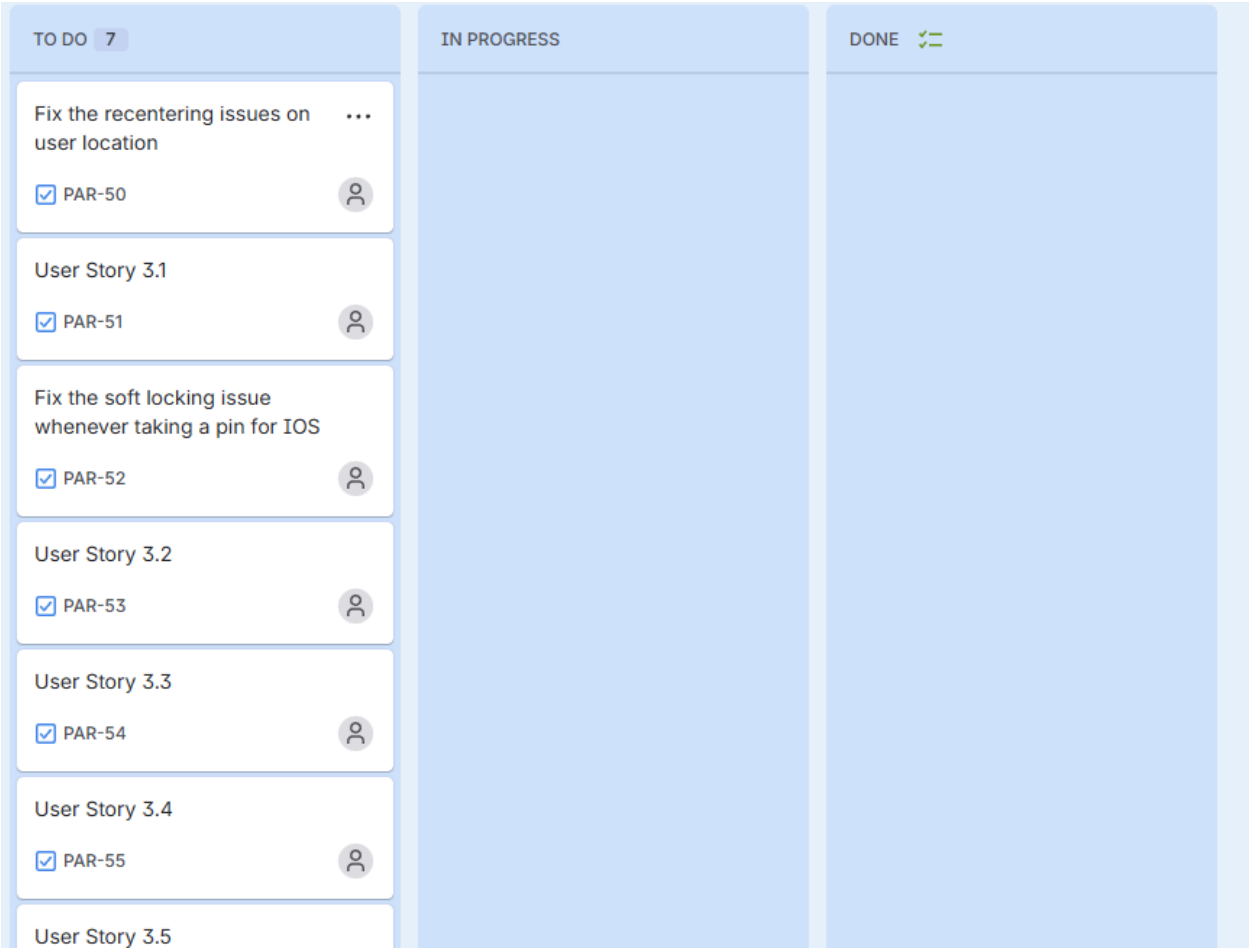
- Kenny: Product Owner
- Ethan: Developer
- Maxwell: Developer
- Kyle: Scrum Master

**Scrum Times:**

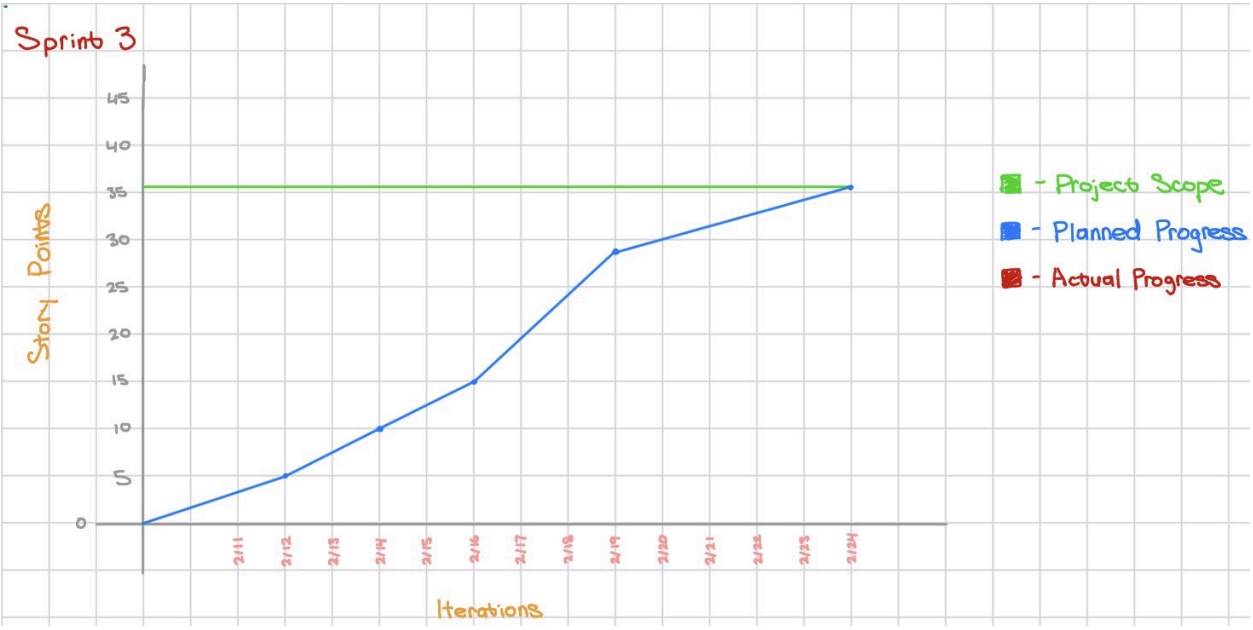
MWF after class

**Scrum Board:**

Sprint 3 Plan, Parking Finder, Parking Finder, 2/24/26, Revision 2, 2/9/26




**Burnup Chart:**



Sprint 3 Plan, Parking Finder, Parking Finder, 2/24/26, Revision 2, 2/9/26